EDUMENTRY AI HACKATHON 1.0

# TutorConnect Nigeria

A full-stack tutor marketplace connecting Nigerian students and parents with tutors


## TEAM MEMBERS

Uchekwe Uchechukwu Hephzibah (Team Leader)

Chioma Helen Ezeufonna (Frontend)

Uzochukwu Mary Cynthia Enyemakachukwu (Tutor Discovery & Booking)

Bodman Ebuka Precous (Backend)

Pius Kaosisochukwu Aminie (Database & Deployment)





Illustrative image: AI-generated photo from the project's public/images folder, not a screenshot.

AGENDA AND SPEAKING ORDER

# Five presenters, one continuous story

| # | Presenter | Role | Assigned topics | Slides | Time |
|---|---|---|---|---|---|
| 1 | **Uchekwe Uchechukwu Hephzibah** | Team Leader | Introduction, problem, aim, users, features | 3–6, 23 | ~3.5 min |
| 2 | **Chioma Helen Ezeufonna** | Frontend Development | Technologies, interface, screens, responsiveness | 7–10 | ~3 min |
| 3 | **Uzochukwu Mary Cynthia Enyemakachukwu** | Tutor Discovery and Booking | Search, profiles, availability, booking workflow | 11–14 | ~3 min |
| 4 | **Bodman Ebuka Precous** | Backend Development | Next.js API routes, authentication, data flow | 15–18 | ~3 min |
| 5 | **Pius Kaosisochukwu Aminie** | Database, Architecture, Deployment | Tables, relationships, deployment, future work, revenue model | 19–22 | ~3 min |

Each presenter opens with a labelled section slide, presents their own slides, and ends with a handover. The Team Leader opens the talk and closes it with the conclusion (slide 23). Total speaking time is about 17 minutes.

# PRESENTER 1

Speaking order 1 of 5

**PRESENTER:**

## Uchekwe Uchechukwu Hephzibah

**ROLE:**

Team Leader

**TOPIC:**

Introduction to TutorConnect Nigeria

**WHAT THIS SECTION COVERS:**

- Project introduction and background
- Problem statement, aim and objectives
- Intended users and major features
- Handover to the technical sections
- Closing summary and questions

PRESENTER 1 · Uchekwe Uchechukwu Hephzibah

# Background and problem statement

Illustrative scenarios, not survey data. They describe the kinds of difficulty the project was designed to reduce.

## A parent searching for help

A parent in Lagos needs a Mathematics tutor for a child preparing for WAEC. Recommendations come from friends, and there is no simple way to compare qualifications, rates, or availability.

## A student with exam pressure

A Senior Secondary student wants sessions after school. Arranging a time usually means many back-and-forth phone messages before anything is confirmed.

## A tutor with no reach

A qualified tutor depends on personal networks to find students, so good teachers outside those networks are hard to discover.

## Trust is hard to judge

Families have little way to tell whether a tutor's identity or academic credentials have been checked before they agree to pay.

PRESENTER 1 · Uchekwe Uchechukwu Hephzibah

# Aim, objectives, and project overview

## AIM

**Make it simpler and more transparent to find and book a tutor in Nigeria.**

- Show verified tutor profiles
- Support search and filters by subject, budget, and availability
- Turn a booking into a clear request with a status
- Collect ratings and reviews after completed sessions

## INTENDED USERS

**Students:** Find tutors and manage sessions

**Parents:** Search for a child's tutor and track bookings

**Tutors:** Publish a profile, set availability, accept requests

**Administrators:** Review credentials and approve or suspend accounts

## MAJOR PLATFORM FEATURES (as built in the repository)

| | | | |
|---|---|---|---|
| Two-mode homepage: Discover & Learn / Teach and Earn | Subject grid and live tutor filters | Public tutor profiles with credentials and reviews | Weekly availability calendar on each profile |
| Booking request modal with price estimate | Student, parent, tutor, and admin dashboards | Messaging between students/parents and tutors | Credential upload and admin verification queue |

PRESENTER 1 · Uchekwe Uchechukwu Hephzibah

# Handover to the frontend section

### THIS SECTION SO FAR

- Introduced TutorConnect Nigeria and its purpose
- Explained the problem families and tutors face
- Stated the aim, objectives, intended users, and features

NEXT: PRESENTER 2

## Chioma Helen Ezeufonna

### Frontend Development

- Frontend development in this project
- Frontend technologies and what each one does
- How the interface is organised for students, parents, tutors, and admins
- Real screenshots of the homepage, tutor search, and dashboards

# PRESENTER 2

Speaking order 2 of 5

**PRESENTER:**

## Chioma Helen Ezeufonna

**ROLE:**

Frontend Development

**TOPIC:**

Frontend Development and User Interface Design

**WHAT THIS SECTION COVERS:**

- What frontend development means here
- Technologies used for the interface
- How the interface is organised for each user
- Homepage, navigation and key screens
- Responsive design and user experience

PRESENTER 2 · Chioma Helen Ezeufonna

# Frontend technologies and interface design

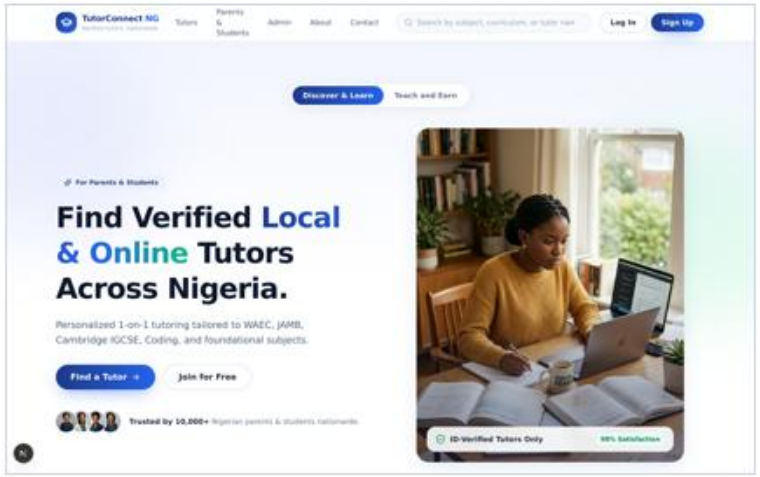
Frontend = everything the user sees and clicks in the browser.

| Technology | Role in this project | Where to find it |
|---|---|---|
| **Next.js 16 (App Router)** | Builds each page from a folder path, such as /tutors or /dashboard/student | app/ folder |
| **React 19** | Builds reusable pieces of the screen: navbar, tutor card, booking modal | components/ folder |
| **TypeScript** | Defines the shape of data, such as a Tutor or a Booking, to catch mistakes early | lib/types.ts |
| **Tailwind CSS v4** | Styles the interface with utility classes and the navy, indigo, and emerald palette | app/globals.css |
| **Framer Motion** | Animates the homepage hero toggle and some transitions | components/HeroSlider.tsx |
| **lucide-react** | Provides the icon set: search, calendar, shield, and others | Navbar, dashboards |
| **next/font (Inter, Plus Jakarta Sans)** | Loads the two typefaces used for body text and headings | app/layout.tsx |

Responsive layout: Tailwind breakpoints change the layout on small screens, and the navbar collapses into a menu button on mobile.

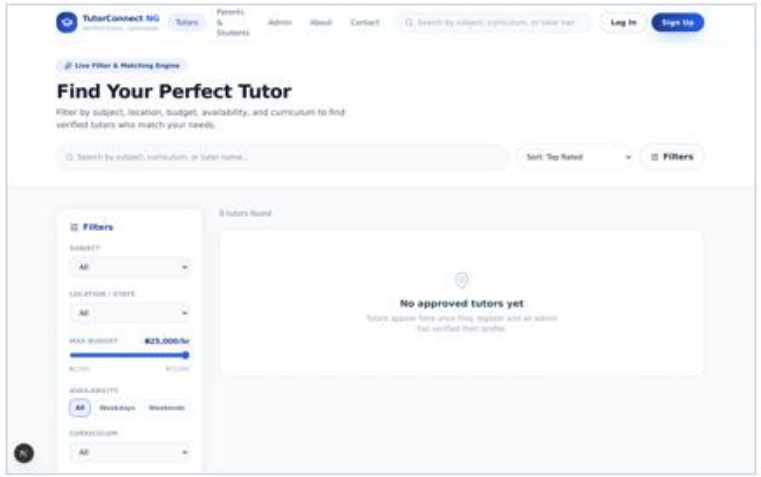
PRESENTER 2 · Chioma Helen Ezeufonna

# TutorConnect Nigeria user interface



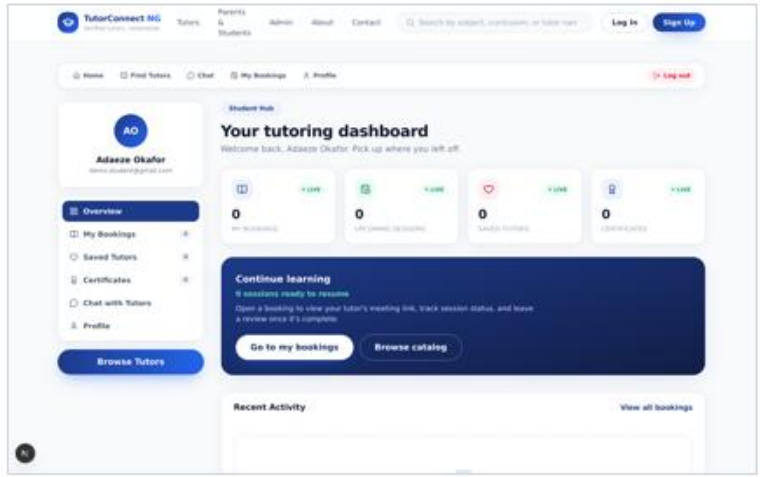

## Homepage hero (real screenshot)

Two modes: Discover & Learn for families, Teach and Earn for tutors.




## Tutor search page (real screenshot)

Shows the empty state. Tutors appear only after database approval.




## Student dashboard (real screenshot)

Sidebar for bookings, saved tutors, chat, and profile. Demo account, no bookings.

- Navigation: Tutors, Parents & Students, Admin, About, Contact, plus Log In and Sign Up.
- Each role has its own dashboard: student/parent, tutor, and admin.
- Empty states are explicit. The app shows a message instead of invented tutors when no approved tutors exist.

PRESENTER 2 · Chioma Helen Ezeufonna

# Handover to tutor discovery and booking

### THIS SECTION SO FAR

- Explained what frontend development means in this project
- Covered the technologies and their roles
- Showed the homepage, navigation, and dashboards
- Explained how the interface supports each user type

NEXT: PRESENTER 3

## Uzochukwu Mary Cynthia Enyemakachukwu

### Tutor Discovery and Booking

- How students and parents find tutors
- Search, filters, and tutor profiles
- Subjects, rates, experience, and availability
- The booking request workflow, step by step

# PRESENTER 3

Speaking order 3 of 5

**PRESENTER:**

## Uzochukwu Mary Cynthia Enyemakachukwu

**ROLE:**

Tutor Discovery and Booking

**TOPIC:**

Tutor Discovery, Profiles, and Booking Workflow

**WHAT THIS SECTION COVERS:**

- How students and parents find tutors
- Search, filters and tutor profiles
- Subjects, rates, experience and availability
- Booking requests and session information
- The booking workflow, step by step

PRESENTER 3 · Uzochukwu Mary Cynthia Enyemakachukwu

# Tutor search and profiles

Students and parents discover tutors, then judge whether a tutor suits them.

### 1 FIND A TUTOR (/tutors)

- Search box: subject, curriculum, or tutor name
- Filters: subject, location / state, maximum budget (₦2,000 to ₦25,000 per hour), availability (all, weekdays, weekends), and curriculum (Nigerian National, British Cambridge, American)
- Verified-only option and sorting (for example, top rated)
- Results come from the database. Only admin-verified tutors are listed.

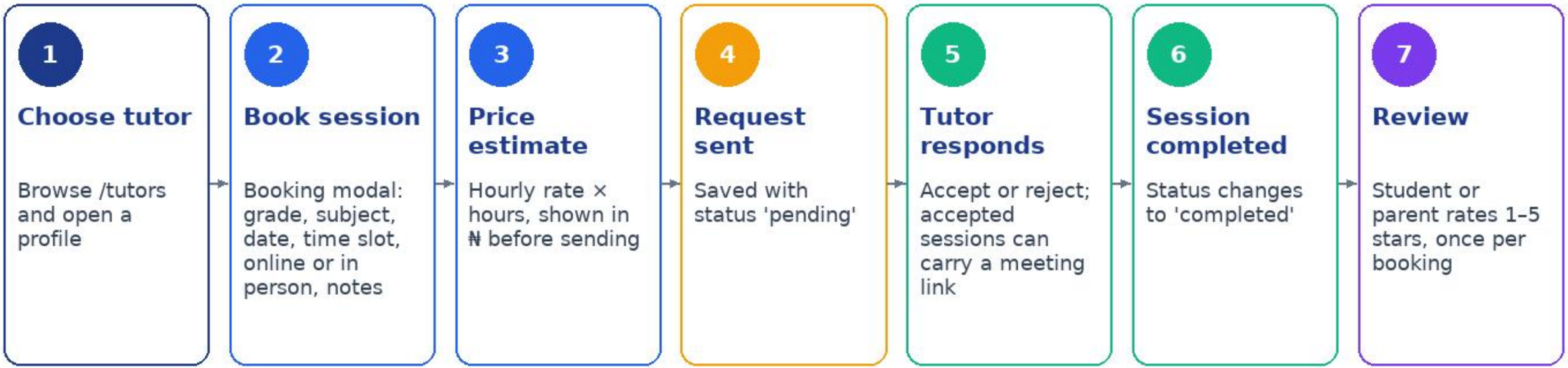
### 2 JUDGE THE PROFILE (/tutors/[id])

- ID-verified badge when the admin has verified the profile
- Headline, bio, years of experience, and curriculum
- Subjects taught and the hourly rate in naira (₦)
- Weekly availability calendar, plus credential status (ID and academic credential uploaded)
- Star rating and written reviews from students

PRESENTER 3 · Uzochukwu Mary Cynthia Enyemakachukwu

# Booking workflow

From a request to a completed session, as implemented in the app.




- Guests can send a request without logging in, and the booking is labelled "Guest Student".
- Only pending requests can be edited. Statuses are pending, accepted, rejected, completed, and cancelled.
- The price is calculated in the browser and saved as sent. Stronger server-side checks are planned (see slide 21).

PRESENTER 3 · Uzochukwu Mary Cynthia Enyemakachukwu

# Handover to the backend

### THIS SECTION SO FAR

- Explained how families discover tutors and judge suitability
- Covered filters, tutor profiles, subjects, rates, and availability
- Walked through the booking request workflow, from request to review

NEXT: PRESENTER 4

## Bodman Ebuka Precous

### Backend Development

- What the backend does
- Next.js API routes and the endpoint map
- Password hashing and JWT-based login
- How the frontend calls the backend, and how data is saved

# PRESENTER 4

Speaking order 4 of 5

**PRESENTER:**

## Bodman Ebuka Precous

**ROLE:**

Backend Development

**TOPIC:**

Backend Development and API Integration

**WHAT THIS SECTION COVERS:**

- What the backend does
- Next.js API routes and the endpoint map
- Password hashing and JWT authentication
- How the frontend calls the backend
- Booking, tutor, review and messaging logic

PRESENTER 4 · Bodman Ebuka Precous

# Backend technologies and API operations

Backend = the code that runs on the server, checks requests, and talks to the database.

| Route | Methods | What it handles |
|---|---|---|
| **/api/tutors** | GET | List approved tutors with filters |
| **/api/auth** | POST, PUT, DELETE, GET | Sign up, log in, profile, children, saved tutors, availability, documents, verification, admin actions |
| **/api/bookings** | GET, POST, PATCH, DELETE | List bookings, create a request, accept/reject or edit, remove |
| **/api/reviews** | GET, POST | Submit a 1–5 star review for a completed booking; list reviews |
| **/api/messages** | GET, POST | Read conversations and send a message between a student/parent and a tutor |

**LOGIN SECURITY**

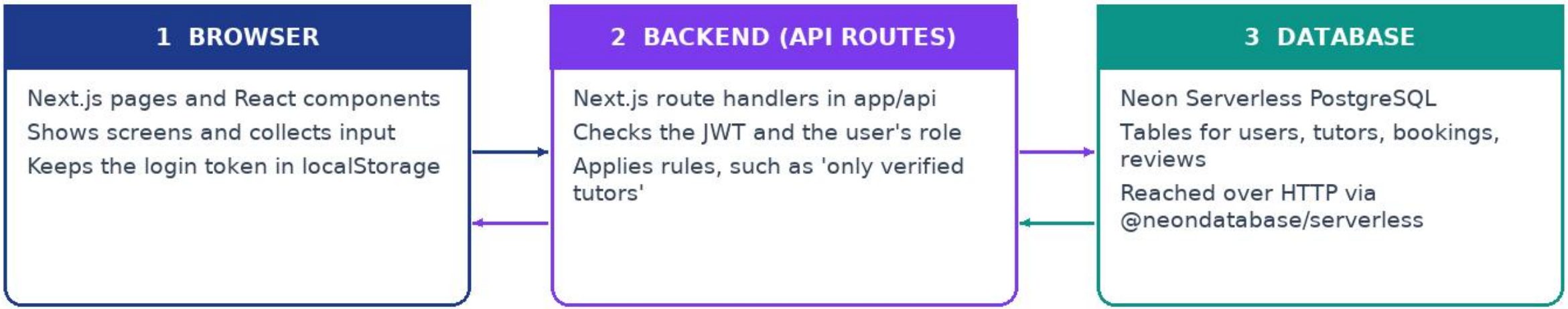
- Passwords are hashed with bcryptjs before saving.
- Successful login returns a JWT, valid for 7 days.
- Protected routes read the token from the Authorization header.

**KNOWN GAPS (not yet built)**

- Booking status changes do not yet check that the caller owns the booking.
- Login does not check account approval status.
- Authentication = proving who you are. Authorisation = what you may do. Both are needed.

PRESENTER 4 · Bodman Ebuka Precous

# How the three layers communicate




Arrows: browser to API = HTTPS request with Bearer token; API to database = SQL query via the Neon driver. Responses flow back the other way.

## One user action, step by step

1. **User clicks 'Send request'**
2. **Browser sends JSON + token**
3. **API verifies token and role**
4. **SQL writes the booking row**
5. **JSON reply; screen updates**

Without DATABASE_URL (demo mode): login and chat use an in-memory store that resets on restart; tutor and booking pages show empty states instead of invented data.

PRESENTER 4 · Bodman Ebuka Precous

# Handover to the database and deployment section

### THIS SECTION SO FAR

- Explained the backend and what the API routes do
- Covered authentication: password hashing and JWT tokens
- Showed how the frontend, backend, and database communicate

NEXT: PRESENTER 5

## Pius Kaosisochukwu Aminie

### Database, Architecture and Deployment

- Why the application needs a database
- The main tables and how they relate
- How the system is deployed on Vercel with Neon
- Limitations, future improvements, and a proposed revenue model

# PRESENTER 5

Speaking order 5 of 5

**PRESENTER:**

## Pius Kaosisochukwu Aminie

**ROLE:**

Database, Architecture and Deployment

**TOPIC:**

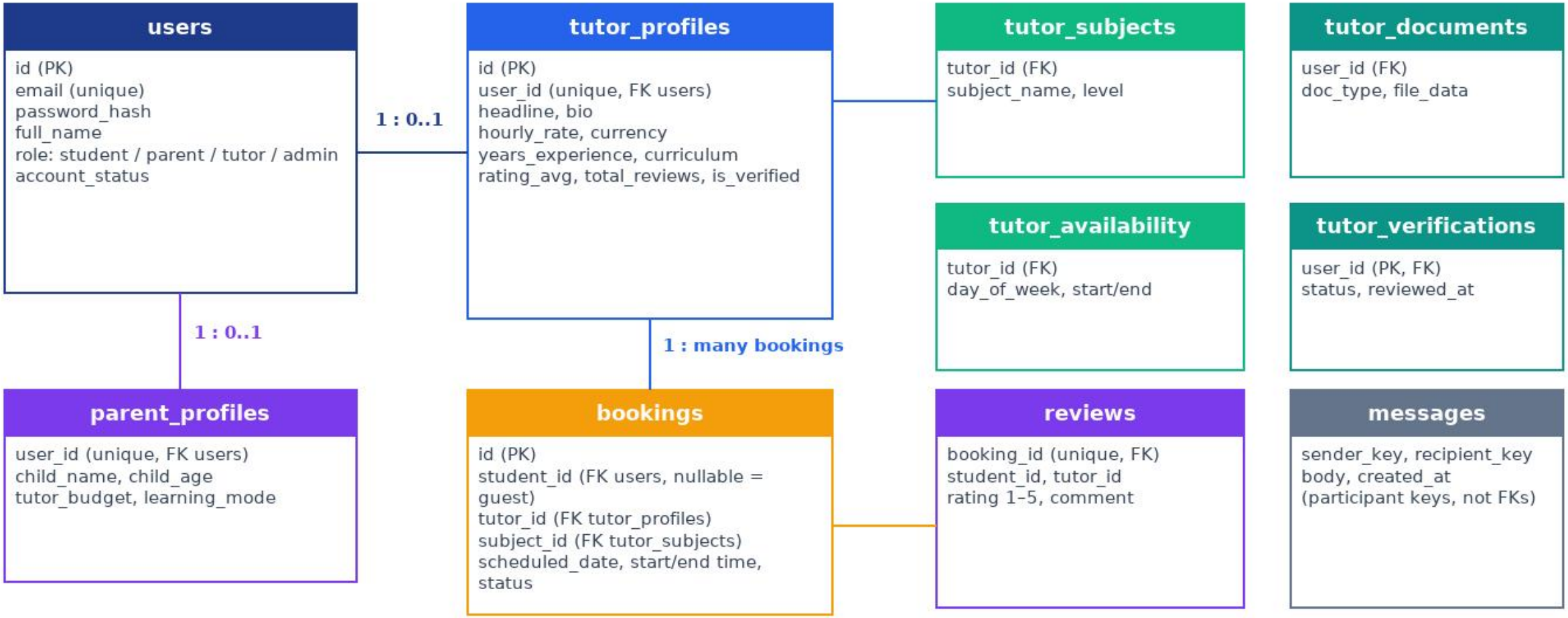
Database, System Architecture, and Deployment

**WHAT THIS SECTION COVERS:**

- Why the application needs a database
- Neon Serverless PostgreSQL and its tables
- How the data is connected
- System architecture and deployment on Vercel
- Limitations, future work and a proposed revenue model

PRESENTER 5 · Pius Kaosisochukwu Aminie

# Database design and relationships

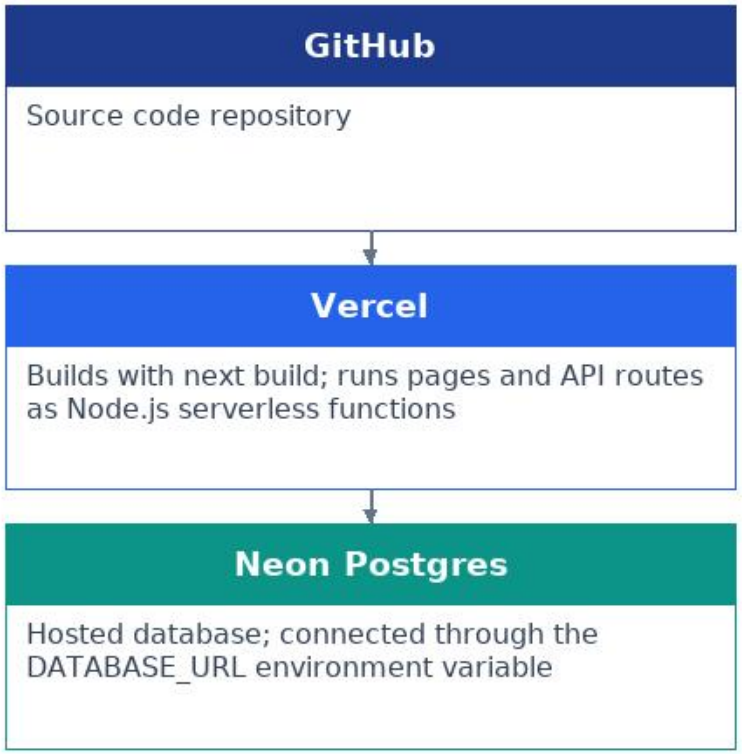



Key: 1 : many = one row links to many rows. FK = foreign key. PK = primary key. Lines show the main links; other keys are listed inside each box.

PRESENTER 5 · Pius Kaosisochukwu Aminie

# Deployment, limitations, and future improvements

## HOW IT IS DEPLOYED




Environment variables are set in Vercel. Names only: DATABASE_URL, JWT_SECRET. Values are never shown.

## LIMITATIONS TODAY

- Dashboards are not protected by server-side middleware; the browser checks the token.
- Booking status changes do not check ownership.
- Price is calculated in the browser and saved as sent.
- Login does not check account approval status.
- Signup accepts Gmail addresses only.
- No payment processing; the app records a price only.
- Homepage statistics and testimonials are static placeholder text, not live data.

## REALISTIC NEXT STEPS

- Add middleware that checks the token and role before dashboards load.
- Check booking ownership on every status change.
- Calculate prices on the server from the tutor's rate.
- Block login for accounts that are not approved.
- Replace static homepage claims with real database counts or remove them.
- Add automated tests for the API routes.

PRESENTER 5 · Pius Kaosisochukwu Aminie

# Proposed revenue model

**PROPOSED FUTURE MODEL. NOT IMPLEMENTED. The current code has no payment gateway, no commission calculation, and no subscriptions.**

## Commission per completed session

A percentage of each booking's total price, taken when a session is marked completed. Needs a payment flow that does not yet exist.

## Featured placement for tutors

Verified tutors could pay to appear higher in search results. Must be clearly labelled so ranking stays fair.

## Premium plan for parents

A paid plan with progress summaries and priority matching. Would need new features that are not built.

## School and centre partnerships

Licensed access for schools or tutoring centres to manage their own tutors and bookings.

PRESENTER 1 · Uchekwe Uchechukwu Hephzibah

# Conclusion, thank you, and questions

## WHAT WE BUILT, IN ONE LOOK

**Introduction:** The problem, our aim, and the users we serve
**Uchekwe Uchechukwu Hephzibah**

**Frontend:** Next.js and React screens for every user type
**Chioma Helen Ezeufonna**

**Discovery and booking:** Search, profiles, and a clear request workflow
**Uzochukwu Mary Cynthia Enyemakachukwu**

**Backend:** API routes, bcrypt hashing, and JWT login
**Bodman Ebuka Precous**

**Database and deployment:** Neon PostgreSQL, Vercel, and a clear list of gaps
**Pius Kaosisochukwu Aminie**

## Thank you

Thank you for listening. Our team is happy to take your questions.

**Questions?**